THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 2: **Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following equals 5.62×10^4? 1

(a) 5620 **(b)** 0.000562 **(c)** 56200 **(d)** 562000

2. $\frac{1}{81}$ expressed as a power of 3 is: 1

(a) 3^{-5} **(b)** 3^{-3} **(c)** 3^{-4} **(d)** 3^4

3. In 3^5, the base is: 1

(a) 15 **(b)** 3 **(c)** 5 **(d)** 243

4. How many zeros are there in one crore written in figures? 1

(a) 6 **(b)** 8 **(c)** 9 **(d)** 7

5. The value of $\frac{1}{5^{-2}}$ is: 1

(a) -25 **(b)** $\frac{1}{25}$ **(c)** 10 **(d)** 25

6. The value of $(-5)^3$ is: 1

(a) 125 **(b)** -125 **(c)** 128 **(d)** 15

7. The standard form of 0.0000052 is: 1

(a) 52×10^{-6} **(b)** 5.2×10^{-6} **(c)** 5.2×10^{-5} **(d)** 5.2×10^{-7}

8. Which of the following equals 9.1×10^{-5}? 1

(a) 0.000091 **(b)** 0.0000091 **(c)** 910000 **(d)** 0.00091

9. $\frac{1}{125}$ expressed as a power of 5 is: 1

(a) 5^3 **(b)** 5^{-4} **(c)** 5^{-3} **(d)** 5^{-2}

10. In 11^2, the base is: 1

(a) 2 **(b)** 22 **(c)** 11 **(d)** 121

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write $\left(\frac{5}{2}\right)^{-2}$ in simplest form. 2

12. The mass of a dust particle is 0.000000753 kg. Express it in standard form. 2

13. Express $(5^2)^3 \times 5^{-2}$ as a single power of 5. 2

14. Write $\left(\frac{4}{3}\right)^{-2}$ in simplest form. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Express each as a single power of 5: 3

(i) $5^2 \times 5^{-4}$

(ii) $(5^3)^{-1}$

(iii) $5^1 \div 5^{-3}$.

16. Find the value of $\left(\frac{5}{4}\right)^{-1} \times \left(\frac{5}{4}\right)^{-2}$. 3

17. Express the following in standard form: 3

(i) 750000000

(ii) 0.0029

(iii) 41000.

OR

Express each as a single power of 7: 3

(i) $7^3 \times 7^{-1}$

(ii) $(7^2)^{-2}$

(iii) $7^2 \div 7^{-2}$.

18. Find the value of $\left(\frac{2}{5}\right)^{-2} \times \left(\frac{2}{5}\right)^{-1}$. 3

OR

Express the following in standard form: 3

(i) 3600000

(ii) 0.000041

(iii) 990000.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Find x if 5

(i) $5^x \times 5^4 = 5^{10}$

(ii) $(5^x)^3 = 5^{18}$

(iii) $5^{x-3} = \frac{1}{5^2}$.

OR

Write 0.009 and 0.00003 in standard form. Then find their product and quotient in standard form. 5

20. Find x if 5

(i) $7^x \times 7^1 = 7^6$

(ii) $(7^x)^2 = 7^{14}$

(iii) $7^{x-5} = \frac{1}{7^3}$.

OR

Write 4500000 and 150 in standard form. Then find their product and quotient in standard form. 5

END OF QUESTION PAPER Total: 40 Marks

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